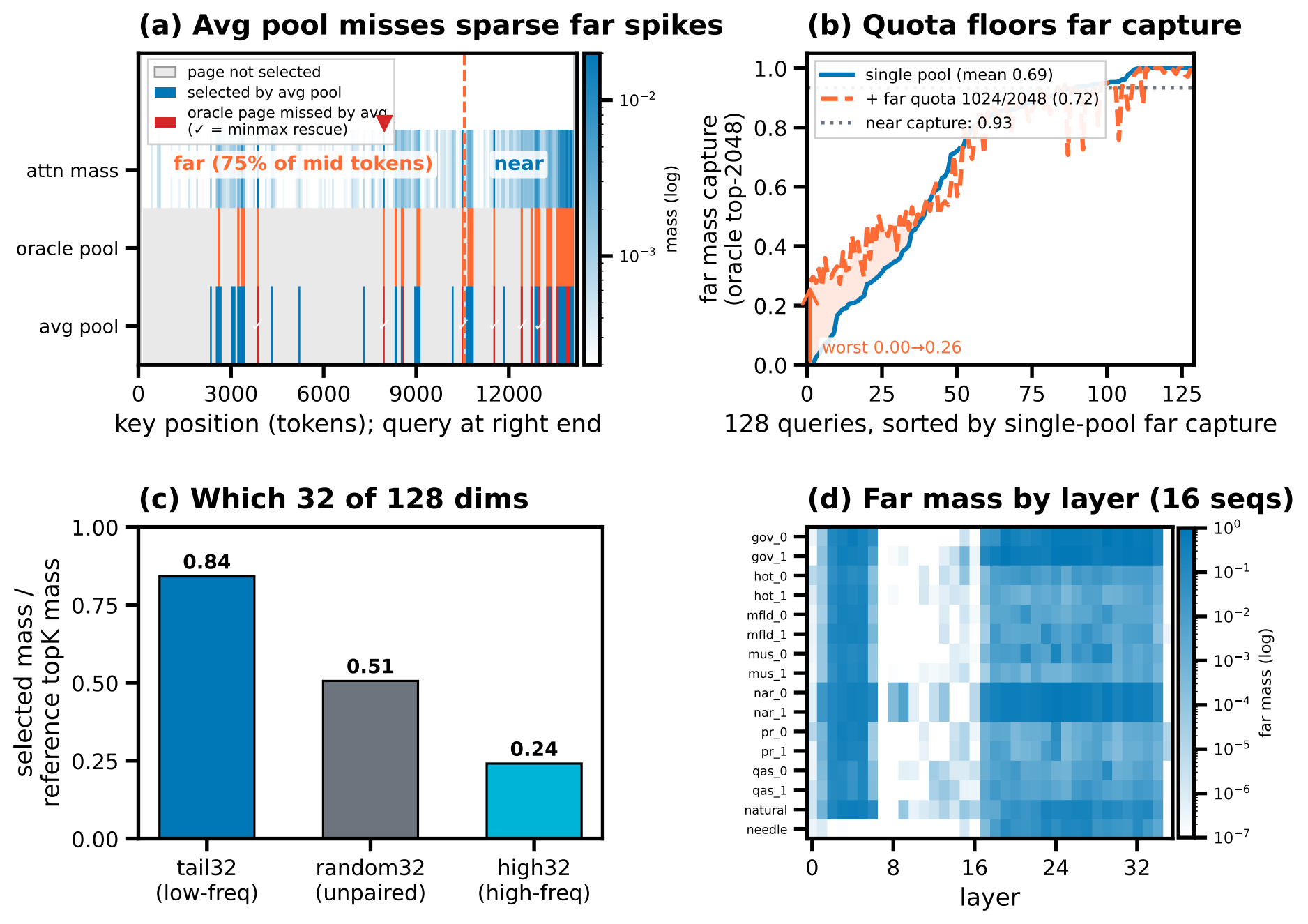
(a) Avg pool misses sparse far spikes
page not selected
selected by avg pool
oracle page missed by avg
(✓ = minmax rescue)
far (75% of mid tokens)
near
attn mass
oracle pool
avg pool
mass (log)
key position (tokens); query at right end
(b) Quota floors far capture
single pool (mean 0.69)
+ far quota 1024/2048 (0.72)
near capture: 0.93
worst 0.00→0.26
far mass capture (oracle top-2048)
128 queries, sorted by single-pool far capture
(c) Which 32 of 128 dims
selected mass / reference topK mass
0.84
0.51
0.24
tail32 (low-freq)
random32 (unpaired)
high32 (high-freq)
(d) Far mass by layer (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
layer
far mass (log)